第 1 章 Maximum Likelihood Estimation

1.1 Likelihood function and the ML principle

Let $f(y;\theta_0)$ denote the probability density function of the random variable y given θ_0. Our objective is to estimate the true parameter vector θ_0. The joint density of n iid observations of y is

$$f(y_1,\dots,y_n|\theta_0)=\prod_{i=1}^{n} f(y_i;\theta_0) \tag{1}$$

Now taking $f(y_i;\theta)$ as a function of θ given $y_1,\dots,y_n$ and write

$$L(\theta|y_1,\dots,y_n)=\prod_{i=1}^{n} f(y_i;\theta) \tag{2}$$

gives the **likelihood function**, the likelihood that the population parameter is θ given the observed sample.

1.1.1 ML principle

The **Maximum Likelihood (ML) principle** suggests that estimators of the unknwon parameters are obtained by **maximizing $L(\theta)$ with respect to θ**. If the random variable y is discrete, $\hat\theta$ yields the value of θ that maximizes the probability of observing the sample that actually occurred.

Usually, $\hat\theta$ can be obtained by solving the likelihood equation

$$\frac{\partial \log L(\theta)}{\partial\theta}|_{\hat\theta}=0 \tag{3}$$

where $\log L(\theta|y_1,\dots,y_n)=\sum_{i=1}^{n}\log f(y_i;\theta)$. Occationally the ML estimator is not unique and $\log L(\theta)$ may have only one global maximum, but multiple local maxima.

The **regularity conditions** are the following:

1. The first three derivatives of $\log f(y|\theta)$ with respect to θ are continuous and finite for almost all y and for all θ.
2. For all values of θ, $|\frac{\partial^3 \log f(y|\theta)}{\partial\theta_j\partial\theta_k\partial\theta_l}|$ is limited by a function that has finite expectations.
3. The domain of y does not depend on θ. For example of the uniform distribution, since the density function depends on θ_0, the theorem does not hold.
4. θ is an interior point to the compact parameter space Θ.

1.1.2 Bartlett Identities

Define the **score vector** $S(\theta)$ and the **Hessian matrix** $H(\theta)$ as

$$S(\theta) = \frac{\partial \log L(\theta)}{\partial \theta} = \sum_{i=1}^{n} \frac{\partial \log f(y_i|\theta)}{\partial \theta} = \sum_{i=1}^{n} s_i(\theta) \tag{4}$$

$$H(\theta) = \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^{n} \frac{\partial^2 \log f(y_i|\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^{n} H_i(\theta) \tag{5}$$

- The **First Bartlett identity** states that $E[s_i(\theta_0)] = 0$. Hence, $E[S(\theta_0)] = 0$.
- The **Second Bartlett identity** states that $Var[s_i(\theta_0)] = -E[H_i(\theta_0)]$.

  证明.

  $$Var\left[\frac{\partial \log f(y_i|\theta)}{\partial \theta}\right] = -E\left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right]$$

  where y_i is a random variable. □

$Var[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)']$ defines the **Fisher's information matrix** denoted as $\mathcal{I}(\theta_0)$. Hence, the result $Var[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)'] = -E[H_i(\theta_0)]$ is also called the **information matrix identity**.

证明.

$$\begin{aligned} Var[s_i(\theta)] &= E[[s_i(\theta_0) - \underbrace{E[s_i(\theta_0)]}_{=0,\text{ First Bartlett}}] \times [s_i(\theta_0) - \underbrace{E[s_i(\theta_0)]}_{=0,\text{ First Bartlett}}]'] \\ &= E[s_i(\theta_0)s_i(\theta)'] \end{aligned}$$

□

1.2 Properties of MLE

Under the assumed regularity conditions of MLE possesses the following properties:

1. **Consistency** meaning $plim\hat{\theta} = \theta_0$
2. **Asymptotic normality** meaning $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$. The goal is to estimate the $[\mathcal{I}(\theta_0)]^{-1}$ matrix (Fisher's information matrix) $\rightarrow$ need to go to a known distribution (normal).
3. **Asymptotic efficiency** meaning that if $\tilde{\theta}$ is a regular consistent asymptotically normal estimator such that $\sqrt{n}(\tilde{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega)$, then $\Omega - [\mathcal{I}(\theta_0)]^{-1}$ is positive semi-definite, i.e., under these RC, the MLE asymptotically achieves the **Cramer-Rao** lower bound which is given by $[\mathcal{I}(\theta_0)]^{-1}$. Information matrix is smaller than Ω in the matrix sense, and the information matrix is the lower bound.
4. **Invariance** meaning that if $c(\theta)$ is a continuous and continuously differentiable one-to-one function, the MLE of $\gamma = c(\theta_0)$ is $c(\hat{\theta})$.

Now, don't forget that θ_0 is not known, so we want to estiamte the information matrix.

1.2.1 Estimators of the information matrix

There are three commonly used estimators of $\mathcal{I}(\theta_0)$:

- **Expected Information**: If the form of the expected values of the second derivatives of the log-likelihood function is known, then we can evaluate $\mathcal{I}(\theta)$ at $\hat{\theta}$. This rarely happens in practice.
- **Observed information**: Simply use $-\hat{H}(\hat{\theta})$.

证明.

$$\begin{aligned}
\mathcal{I}(\theta_0) = Var[s_i(\theta_0)] &= -E[H(\theta_0)] \\
&= Var\left[\frac{\partial \log f(y_i|\theta_0)}{\partial \theta_0}\right] \\
&= -E\left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right] \\
&\underset{\text{avg.}}{=} \frac{1}{N}\sum_{i=1}^{n}\left(\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right) \\
&= -\hat{H}(\hat{\theta})
\end{aligned}$$

□

- **Outer Product of the Gradient (OPG)**: Because of the information matrix identity, we can also use $n^{-1}\sum_{i=1}^{n} s_i(\hat{\theta})s_i(\hat{\theta})'$. The OPG is notorious for its poor performance with finite samples.

证明.

$$\begin{aligned}
Var[s_i(\theta_0)] &= E[s_i(\theta_0)s_i(\theta)'] \\
&= \frac{1}{N}\sum_{i=1}^{N} s_i(\theta_0)s_i(\theta_0)' \\
&= \frac{1}{N}\sum_{i=1}^{N} s_i(\hat{\theta})s_i(\hat{\theta})'
\end{aligned}$$

□

1.3 Regressors

Suppose the joint distribution of y and x depends on α_0, giving

$$f(y, x|\alpha_0) = f(y|x, \alpha_0)g(x|\alpha_0) \tag{6}$$

$$f(y|x, \alpha_0) = \frac{f(y, x, \alpha_0)}{g(x, \alpha_0)} \tag{7}$$

$$\underbrace{f(y, x, \alpha_0)}_{\text{joint}} = \underbrace{f(y|x, \alpha_0)}_{\text{conditional}}\underbrace{g(x, \alpha_0)}_{\text{marginal}} \tag{8}$$

and suppose α_0 can be divided into θ_0 and δ_0, so that with exogeneity of x, $f(y,x|\alpha_0) = f(y_i|x_i,\theta_0)g(x_i|\delta_0)$. For an iid sample $\{(y_i,x_i)\}_{i=1}^n$,

$$\log L(\theta,\delta|y_i,x_i) = \sum_{i=1}^n \log f(y_i,x_i|alpha) = \sum_{i=1}^n \log f(y_i|x_i,\theta) + \sum_{i=1}^n \log g(x_i|\delta) \tag{9}$$

We are under the assumption that we are only interested in θ_0. Now this assumption that we don't care about δ makes our lives easier because

$$\frac{\partial L(\theta,\delta|y_i,x_i)}{\partial\theta}|_{\hat{\theta}} = 0 \Leftrightarrow \frac{\partial \sum_{i=1}^n \log f(y_i|x_i,\theta_i)}{\partial\theta}|_{\hat{\theta}} = 0$$

The part of $g(x,\delta)$ can be ignored now, only focus on the conditional density function. However, now $E[-H_i(\theta_0)] = \mathcal{I}(\theta_0)$.

1.4 Interpretation and inference in misspecified models

If the likelihood function is **misspecified**, the MLE is generally **inconsistent** for the parameters of interest. However, under the general conditions, $plim\hat{\theta} = \theta^*$, where the **pseudo-true value** θ^* minimizes the Kullback-Leibler divergence where $f_0(y)$ is the true distribution of data.

> **定义 1.4.1.** ***Kullback-Leibner divergence****, also called relative entropy, is a measure of how one probability distribution is different from a second, reference probability distribution.*

This means that the MLE leads to the best approximation in the Kullback-Leibner sense to the true density. However, because Fisher's information matrix identity does not hold, the asymptotic covariance matrix is given by:

$$A^{-1}BA^{-1} \quad A = E[-H_i(\theta^*)] \quad B = E[s_i(\theta^*)s_i(\theta^*)'] \tag{10}$$

$f(y|\theta^*)$ is the best approximation given the wrong space but is still wrong.

1.5 Hypothesis testing

Considering a general set of restrictions to be tested

$$H_0 : h(\theta_0) = 0 \tag{11}$$

where

- θ: vector of parameters in the model and the true value is given by θ_0.
- $h(\theta)$ is a $d \times 1$ vector of restrictions.
- $\hat{\theta}$ is the unrestricted MLE where its value maximizes $L(\theta)$.
- $\tilde{\theta}$ is the restricted MLE where its value maximizes $L(\theta)$ and satisfies $h(\theta) = 0$.

1.5.1 The Wald Test

Compares $h(\hat{\theta})$ with 0 since $h(\tilde{\theta}) = 0$. The test statistic is

$$\mathcal{W} = n \times h(\hat{\theta})' \left[G(\hat{\theta})' [\mathcal{I}(\hat{\theta})]^{-1} G(\hat{\theta}) \right]^{-1} h(\hat{\theta}) \tag{12}$$

where $G(\theta) = \frac{\partial h(\theta)}{\partial \theta}$.

Under the null hypothesis,

$$\mathcal{W} \xrightarrow{d} \mathcal{X}^2(d) \tag{13}$$

Wald test is not invariant to how restrictions are formulated. Equivalent non-linear ($\beta_0/(1-\alpha_0) = 1$) and linear ($\beta_0 + \alpha_0 - 1 = 0$) restriction may lead to different values of $\mathcal{W}$.

1.5.2 The Likelihood Ratio Test

Compare $L(\hat{\theta})$ and $L(\tilde{\theta})$. If $h(\theta_0) = 0$ then $L(\hat{\theta})$ should be close to $L(\tilde{\theta})$. The test is base on the likelihood ratio $\lambda = \frac{\mathcal{L}(\tilde{\theta})}{\mathcal{L}(\hat{\theta})}$ with the test statistic

$$\mathcal{LR} = -2\log(\lambda) = 2\{\log \mathcal{L}(\hat{\theta}) - \log \mathcal{L}(\tilde{\theta})\} \tag{14}$$

We may encounter convergence problem when maximizing.

Under the null hypothesis,

$$\mathcal{LR} \xrightarrow{d} \mathcal{X}^2(d) \tag{15}$$

1.5.3 The Langrage Multiplier or Score Test

Compare $S(\tilde{\theta})$ with 0 since $S(\hat{\theta}) = 0$. The test statistic is

$$\mathcal{LM} = S(\tilde{\theta})' [\mathcal{I}(\hat{\theta})]^{-1} S(\tilde{\theta}) \frac{1}{n} \tag{16}$$

Under the null hypothesis,

$$\mathcal{LM} \xrightarrow{d} \mathcal{X}^2(d) \tag{17}$$

The LM test works best in real life application. It is under weaker regulated conditions but it's still difficult to compute the $S(\theta)$ function.

第 2 章 Binary Choice Models

2.1 Linear probability model

In many applications, the variate of interest is binary, i.e., takes values 0 and 1. Let's denote $X = (X_1, \dots, X_k)'$ and the objective of the regression model is to estimate $E[Y|X]$ where $E[Y|X] = P(Y = 1|X)$.

In the **linear probability model** we assume that

$$P(Y = 1|X) = \beta_1 + \beta_2 X_2 + \cdots + \beta_k X_k \tag{18}$$

thus the **interpreation of β_j** is the **change in the probability of success when x_j changes**.

$$\frac{\partial P(Y = 1|X)}{\partial X_j} = \beta_j, \quad j = 2, \dots, k \tag{19}$$

and the **predicted Y** is the **predicted probability of success**.

The linear probability model is estimated using **OLS**. But there is the potential problem that the fitted values can be outside $[0, 1]$. Even without predictions outside $[0, 1]$, we may estimate effects that imply a change in x changes the probability by more than $+1$ or -1. $\hat{\beta}$ too big that would not make sense. Also, this model **violates the assumption of homoskedasticity**, therefore we should use the Eicker-Huber-White **robust standard errors** to make inference. Homoskedasticity states that $\text{Var}[Y|X]$ does not vary with x but here it necessarity depends on x.

证明.

$$\text{Var}[Y|X] = \underbrace{E[Y^2|X]}_{(1)} - E[Y|X]^2$$

(1):

$$E[Y^2|X] = 1^2 \times P(Y = 1|X) + 0^2 \times P(Y = 0|X) = P(Y = 1|X)$$

Now we have,

$$\text{Var}[Y|X] = P(Y = 1|X) - P(Y = 1|X)^2 = \underbrace{P(Y = 1|X)}_{\beta_1 + \beta_2 x_2 + \cdots + \beta_k x_k} [1 - P(Y = 1|X)]$$

So we see that the variance depends on the regressors. □

2.2 Index models for binary response

The alternative to the linear model is to assume that $E[Y|X] = P(Y=1|X) = G(X'\beta_0)$, where $G(\cdot)$ is bounded between 0 and 1. Thus, we have,

$$Y = \begin{cases} 1 & \text{with probability } G(X'\beta_0) \\ 0 & \text{with probability } 1 - G(X'\beta_0) \end{cases} \tag{20}$$

with the log-likelihood function of

$$\log\{L(\beta)\} = \sum_{i=1}^{n} Y_i \log(G(X_i'\beta)) + \sum_{i=0}^{n} (1 - Y_i) \log(1 - G(X_i'\beta)) \tag{21}$$

The **MLE** should be used to find the estimators, $\hat{\beta}_{ML}$.

Note that it is a system of non-linear equations to be solved to find $\hat{\beta}_{ML}$ thus we have to resort to numerical methods. There is no closed form solution for this estimator. **Consistency and asymptotic normality** follow the general results described for MLE under regulairty conditions. Essentially, the **main requirement for consistency** is $E[Y|X] = P(Y=1|X) = G(X'\beta_0)$.

In a correctly specified model, $\hat{\beta}$ is consistent and asymptotically normally distributed with variance-covariance matrix $[\mathcal{I}(\beta_0)]^{-1}$,

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(0, [\mathcal{I}(\beta_0)]^{-1})$$

Inference is done using the Wald, Likelihood Ratio, and Lagrange multiplier statistics as usual.

2.2.1 $G(X'\beta_0)$ Choices

The most popular choices are

- The **Logit Model**

$$G(X'\beta_0) = \frac{\exp(X'\beta_0)}{1 + \exp(X'\beta_0)} \tag{22}$$

- The **Probit Model**

$$G(X'\beta_0) = \Phi(X'\beta_0) \tag{23}$$

  where $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} \exp(-\frac{u^2}{2})\, du$ is the Standard Normal Distribution function.

Both models are non-linear and require **MLE**. Other possible models are

- The **Log-Weibull** Distribution

$$G(X'\beta_0) = \exp(-\exp(X'\beta_0))$$

- The **Complementary log-log Model**, Gompertz Distribution

$$G(X'\beta_0) = 1 - \exp(-\exp(X'\beta_0))$$

- $$G(X'\beta_0) = \Phi(X'\beta_0)^\tau$$

  where τ needs to be estimated.

- $$G(X'\beta_0) = 1 - \exp(-\omega \exp(X'\beta_0))^{-\frac{1}{\omega}}$$

  where ω needs to be estimated. Note that for $\omega = 1$ we have the Logit model, and $\lim_{\omega\to 0} G(X'\beta_0) = 1 - \exp(-\exp(X'\beta_0))$.

Estimating additional parameters of the distribution allows the data to tell us the true model which is a major benefit.

In statistics, a more common interpretation of the coefficients is in terms of **marginal effects** on the odds ratio rather than on the probability. $\frac{P(Y=1|X)}{(1-P(Y=1|X))}$ measures the **probability of yes relative to no** and is called the **odds ratio** or relative risk. For the logit model, the log-odds ratio is linear in the regressors.

证明.

$$\begin{aligned}
P(Y=1|X) &= \frac{\exp(X'\beta_0)}{1+\exp(X'\beta_0)} \Rightarrow \\
\frac{P(Y=1|X)}{1-P(Y=1|X)} &= \frac{\frac{\exp(X'\beta_0)}{1+\exp(X'\beta_0)}}{1-\frac{\exp(X'\beta_0)}{1+\exp(X'\beta_0)}} = \exp(X'\beta_0) \Rightarrow \\
\log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) &= X'\beta_0
\end{aligned}$$

□

2.2.2 Latent Variable Threshold (LVT) Model

Define a latent variable

$$Y^* = X'\beta_0 + \epsilon_t \tag{24}$$

where Y^* is unobserved, thus a **latent variable**. Further assume that ϵ independent of X, $E[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$ with a distribution function of $F(\cdot)$. The observation of Y is given by

$$Y = \begin{cases} 1 & \text{if } Y^* > \lambda \\ 0 & \text{if } Y^* \leq \lambda \end{cases} \tag{25}$$

where λ is the threshold.

The probability of choosing the option is thus given by,

$$
\begin{aligned}
P(Y=1|X) = P(Y^*>\lambda|X) &= P(X'\beta_0+\epsilon>\lambda|X) \\
&= P(\epsilon > -X'\beta_0+\lambda|X) \\
&= 1-P(\epsilon\leq -X'\beta_0+\lambda|X) \\
&\text{independence between } \epsilon, X \\
&= 1-\underbrace{F(-X'\beta_0+\lambda)}_{\text{cumulative distribution function}} \\
&= G(X'\beta_0)
\end{aligned}
$$

with $G(z)=1-F(-z+\lambda)$.

First identification problem:

$$P(Y=1|X)=1-F(-X'\beta_0+\lambda) \tag{26}$$

Notice that if $X_1=1$ meaning that **there is an intercept in the model**, it is not possible to estimate $-\beta_1$ and λ at the intercept separately. Solution is to **set** $\lambda=0$. If $\lambda=0$, and ϵ has a symmetric distribution around 0, then $G(z)=F(z)$ as in this case $1-F(-z)=F(z)$.

Second identification problem:

$$\frac{Y^*}{a}=X'\beta_0^*+\frac{\epsilon}{a} \tag{27}$$

If we divide Y^* from (24) by a constant a and $a>0$, notice that the definition of the observable variable Y (25) does not change. Changing the scale $\rightarrow$ changing of variance but the Y does not change $\Rightarrow$ we cannot identify the scale of Y^*. This implies that **we cannot identify the variance of** ϵ. Error term variance not observed and cannot be identified and estimated. For given β_0^*, value of β_0 depends on a, meaning β_0 is identified up to a scale factor. The solution is to **normalize distribution of** ϵ by fixing σ^2 at a given number $\Rightarrow$ assume σ^2 is known. Normally set to 1 in practice.

2.2.3 Random Utility Models

Suppose an individual has to choose between alternatives a and b with utilities U^a and U^b which are unobservable. But the characteristics of the observations are observable thus we have,

$$
\begin{aligned}
U^a &= X'\beta_a+u_a \\
U^b &= X'\beta_b+u_b
\end{aligned}
$$

We also observe the chosen alternative, say a, which we indicate this choice as $Y = 1$. Then we know that,

$$\begin{aligned} P(Y = 1|X) &= P(U^a > U^b|X) = P(X'\beta_a + u_a > X'\beta_b + u_b) \\ &= P(u_a - u_b > X'(\beta_b - \beta_a)|X) \\ &= P(\epsilon > -X'\beta_0|X) = 1 - F(-X'\beta_0) \end{aligned}$$

with $\epsilon = u_a - u_b$ and $\beta_0 = \beta_a - \beta_b$.

2.3 Marginal effects and average partial effects

It is important to note that the parameters of the model, like those of any nonlinear regression model, are not necessarily the marginal effects. In general, the partial effect associated with variable k, X_{ik} is given by,

$$\Delta E[Y_i|X_i] \equiv \Delta P(Y_i = 1|X_i) \simeq \frac{\partial E[Y_i|X_i]}{\partial X_{ik}} \Delta X_{ik} = \frac{\partial G(X_i'\beta_0)}{\partial X_{ik}} \Delta X_{ik} \tag{28}$$

Let $\omega = X_i'\beta_0 = \beta_1 X_{i2} + \cdots + \beta_k X_{iK}$, then,

$$\frac{\partial E[Y_i|X_i]}{\partial X_{ik}} = \frac{\partial G(\omega)}{\partial(\omega)} \frac{\partial(\omega)}{\partial X_{ik}} = g(\omega)\beta_k = g(X_i'\beta_0)\beta_k \tag{29}$$

where $g(\omega) = \frac{\partial G(\omega)}{\partial \omega}$. Thus, we need to evaluate

$$\Delta e[Y_i|X_i'\beta_{0i}] \equiv \Delta P(Y_I = 1|X_i) \simeq g(X_i'\beta_0)\beta_k \Delta X_{ik} \tag{30}$$

Now, to **compute marginal effects** of variable X_k assuming that **X_{ik} is continuous**, we have options

- Marginal effect at a **particular value** $X_i = X_0$

  $$g(X_0'\hat{\beta}_{ML})\hat{\beta}_{ML,i}\Delta X_{ik}$$

  The most common choices of X_0 are the vector of sample means. This does not make sense if some of the regressors are dummy variables. Or the vector of sample medians.

- Alternative is to compute **Average Partial Effects** which is the sampel average of the individual marginal effects,

  $$\frac{1}{n}\sum_{i=1}^{n} g(X_i'\hat{\beta}_{ML})\hat{\beta}_{ML,k}\Delta X_{ik}$$

The options to compute the marginal effects on a **dummy variable** d, we have the same options

- Marginal effect at a **particular value** $X_i = X_0$

  $$P(Y_i = \widehat{1|X_0}, d_i = 1) - P(Y_i = \widehat{1|X_0}, d_i = 0)$$

  Note that here X_0 denotes all variables in the model at the particular value X_0 except d.

- Average partial effect is thus given by

  $$\frac{1}{n}\sum_{i=1}^{n} P(Y_i = \widehat{1|X_i}, d_i = 1) - \frac{1}{n}\sum_{i=1}^{n} P(Y_i = \widehat{1|X_i}, d_i = 0)$$

  Note that here X_i denotes all variables in the model except d.

2.4 Simple specification tests

2.4.1 Misspecification

If $G(\cdot)$ is **misspecified**, then $\hat{\beta}_{ML}$ is **inconsistent**. Perform **RESET-type test** by testing $H_0 : \delta_1 = \delta_2 = 0$ in the model

$$E[Y_i|X_i] = G(X_i'\beta_0 + \delta_1(X_i'\hat{\beta}_{ML})^2 + \delta_2(X_i'\hat{\beta}_{ML})^3), \quad i = 1, \ldots, n$$

This is a normality test in the probit if $\epsilon_i \sim \mathcal{N}(0,1) \to f(\cdot) = \Phi(\cdot)$. $E[Y_i|X_i] = X_i'\beta$ estimated by OLS and the estimator $\hat{\beta}$ and the model $Y_i = X_i'\beta + \delta_0(X_i'\beta)^2 + \delta_1(X_i'\beta)^3$ and test $H_0 : \delta_0 = 0$ and $\delta_1 = 0$, if we reject $H_0 \to$ model is misspecified.

The RESET test can be tested against mroe general parametric specifications which include additional shape parameters. Consider $G(\cdot) = \Phi(\cdot)^\tau$, use the score statistic to test $H_0 : \tau = 1$, Probit. Or, consider $G(\cdot) = 1 - (1 + \omega \exp(\cdot))^{-\frac{1}{\omega}}$, use the score statistic to test $H_0 : \omega = 1$, Logit.

2.4.2 Heteroskedasticity

In the LVT model, heteroskedasticity leads to misspecification of the conditional mean of Y.

证明. Proof of misspecification for LVT.
The goal is to show that $E[Y|X] \neq G(X'\beta_0)$. As previously shown, $G(a) = 1 - F(a)$.

$$\begin{aligned}
P(Y=1|X) &= P(Y^* > 0|X) = P(X'\beta_0 + k\,h(Z'\gamma_0)\epsilon > 0|X) \\
&= P\left(\epsilon > \frac{-X'\beta_0}{k\,h(Z'\gamma_0)}|X\right) \\
&\text{notice } X' \text{ and condition on } X\text{, we can drop the condition} \\
&= P\left(\epsilon > \frac{-X'\beta_0}{k\,h(Z'\gamma_0)}\right) \\
&= 1 - P\left(\epsilon \leq \frac{-X'\beta_0}{k\,h(Z'\gamma_0)}\right) \\
&= 1 - F\left(\epsilon \leq \frac{-X'\beta_0}{k\,h(Z'\gamma_0)}\right) \\
&= G\left(\epsilon \leq \frac{-X'\beta_0}{k\,h(Z'\gamma_0)}\right) \neq G(X'\beta_0) \to \text{misspecified}
\end{aligned}$$

□

Consider a latent variable,

$$Y^* = X'\beta_0 + k \times h(Z'\gamma_0)\epsilon \tag{31}$$

under the assumptions of ϵ independent of X, $E[\epsilon] = 0$ and $\mathrm{Var}(\epsilon) = 1$ and distirbution function $F(\cdot)$, Z are a vector function of X of size d and h any function with $h > 0, h(0) = 1, h'(0) \neq 0$.

证明. Proof of heteroskedasticity.

The idea is to prove that the variance of the error term is not constant but a function of X.

$$u = k \times h(Z'\gamma_0)\epsilon$$
$$e[u|X] = E[k \times h(Z'\gamma_0)\epsilon|X] = k \times h(Z'\gamma_0)\underbrace{E[\epsilon|X]}_{=0} = 0$$
$$\text{Var}(u|X) = \text{Var}(k \times h(Z'\gamma_0)\epsilon|X) = k^2[h(Z'\gamma_0)]^2\underbrace{\text{Var}(\epsilon|X)}_{=1} = k^2[h(Z'\gamma_0)]^2 \rightarrow \text{Heteroskedasticity}$$

□

We can construct a **LM test** based on the generalized residuals to test $H_0 : \gamma_0 = 0$ which implies **homoskedasticity**. The LM test statistic is calculated as

$$\xi_{LM} = \iota' S(S'S)^{-1}S'\iota \sim \chi^2(d)$$

This is asymptotically equivalent to a **Wald test** with $H_0 : \gamma_0 = 0$ in the model

$$E[Y_i|X_i] = G(X'\beta_0 + (X_i'\hat{\beta}_{ML})Z_i'\gamma_0), \quad i = 1, \ldots, n$$

2.4.3 Goodness of fit

Unlike the linear probaility model, we cannot compute an R^2 to judge the goodness of fit. One possibility is to use the **pseudo** R^2 based on the log likelihood. We can also look at the percent correctly predicted. We can just use the percentage of correct predictions and we can adjust the threshold. This is more for machine learning when prediction matters. For econometrics, the interpretation of β is more important, thus the specification test is more important.

第 3 章 Ordered Data and Count Data Models

3.1 Ordered data

In some problems, the variate of interest assumes more than two discrete outcomes, but these are inherently **orderded**. Ordered data is a generalization of the binary choice model but with an order. Zavoina and McElvey (1975) modeled ordered data using the following latent variable framework,

$$Y_i^* = X_i'\beta_0 + u_i, \quad Y_i = \begin{cases} 0 & Y_i^* \le \mu_0 \\ 1 & \mu_0 < Y_i^* \le \mu_1 \\ 2 & \mu_1 < Y_i^* \le \mu_2 \\ \vdots & \vdots \\ J-1 & \mu_{J-2} < Y_i^* \le \mu_{J-1} \\ J & \mu_{J-1} < Y_i^* \end{cases} \tag{32}$$

where the threshold parameters are such that $0 = \mu_0 < \mu_1 < \cdots < \mu_{J-1}$ and Y_i^* is the latent variable.

If **the distribution of u_i is specified**, the **unknwon parameters** β and $\mu_2, \ldots, \mu_{J-1}$ can be estimated by **ML**. Assume that u_i has dsitribution function $F(\cdot)$ and is independent of X_i, notice that,

$$\begin{aligned} P_0(X_i, \beta_0) &= P(Y_i = 0|X_i) = F(-X_i'\beta_0) \\ P_1(X_i, \beta_0) &= P(Y_i = 1|X_i) = F(\mu_1 - X_i'\beta_0) - F(-X_i'\beta_0) \\ &\vdots \\ P_j(X_i, \beta_0) &= P(Y_i = j|X_i) = F(\mu_j - X_i'\beta_0) - F(\mu_{j-1} - X_i'\beta_0) \\ &\vdots \\ P_J(X_i, \beta_0) &= P(Y_i = J|X_i) = 1 - F(\mu_{J-1} - X_i'\beta_0) \end{aligned}$$

Therefore the log-likelihood function is,

$$\log L(\theta) = \sum_{i=1}^{n} \sum_{i=1}^{J} 1(Y_i = j) \log(P_j(X_i, \beta))$$

As in all discrete choice models, **the variance of u_i is not identified**. And the **ordered-probit and ordered-logit** are the most used special cases of the model.

Interpreting coefficients requires some care and is messy. But, the OP and OL models allow us to obtain the **sign of the partial effects** on $P(Y > j|X_i)$. For a continuous variable x_h for the OP model,

$$\frac{\partial P(Y_i > j|X_i)}{\partial x_h} = \beta_h \Phi(\mu_j - X_i'\beta)$$

If $\beta_h > 0$**, an increase in** x_h **increases the probability that** Y_i **is greater than any value j**. Of course, we can interpret the sign of the parameters in the latent variable model. It is actually easier to interpret the β's of the latent equation to avoid the complications of finding the marginal effect.

A closely related model can be used for **grouped data**. In this case, the threshold parameters are the limits of the intervals. The main difference is that for $J > 0$, **the variance of** u_i **is identified** because the thresholds give information on the scale of u_i. Only need to estimate the β's becasue the μ's are known.

3.2 Count data models

In many applications, the variate of interest is the count of the number of occurences of some event in a given period of time. These data have some very specific characteristics: discreteness, non-negative, and many zeros and a long right-hand tail.

In those conditions, the **standard linear models are not appropriate** because:

- The conditional expectation is necessarily non-negative. The mean is always positive.
- The data is intrinsically heteroskedastic.
- Do not allow the computation of the probability of events of interest.

3.2.1 The Poisson Regression Model

The **Poisson regression model** is defined by,

$$P(Y_i = j|X_i) = \frac{\exp\{-\lambda(X_i.\beta_0)\}\lambda(X_i,\beta_0)^j}{j!}, \quad j = 0, 1, 2, \dots \tag{33}$$

with

$$E[Y_i|X_i] = \text{Var}(Y_i|X_i) = \lambda(X_i, \beta_0) \quad \text{equidispersion assumption}$$
$$\text{Var}(Y_i) = E_x[\lambda(X_i, \beta_0)] + \text{Var}[\lambda(X_i, \beta_0)] \geq E_x[\lambda(X_i, \beta_0)] = E[Y_i]$$

where in general the specification of $\lambda(X_i, \beta_0) = \exp(X_i'\beta_0)$ is adopted. Because $\exp(X_i'\beta_0) > 0$. Therefore,

$$\frac{\partial E[Y_i|X_i]}{\partial X_i} = \exp(X_i'\beta_0)\beta_0$$

and **ML** estimation of β_0 is straightforward.

证明. Proof of $\text{Var}(Y_i)$

$$
\begin{aligned}
\text{Var}(Y_i) & \quad \text{using law of total variance} \\
&= \overbrace{\text{Var}_{X_i}(\underbrace{E[Y_i|X_i]}_{\lambda(\cdot)})}^{\geq 0} + E_{X_i}[\underbrace{E[Y_i|X_i]}_{\lambda(\cdot)}] \Rightarrow \\
\text{Var}(Y_i) &\geq E_{X_i}[\text{Var}(Y_i|X_i)] = E_{X_i}[\lambda(X_i, \beta_0)] \\
&= E_{X_i}[E[Y_i|X_i]] \quad \text{using the law of interated expectations} \\
&= E[Y_i]
\end{aligned}
$$

□

The log-likelihood function is given by,

$$\log L(\beta) = \sum_{i=1}^{n}[-\exp(X_i'\beta) + (X_i'\beta)Y_i - \log(Y_i!)]$$

The **Hessian is negative definite** for all X and β which facilitates the estimation and ensures the unqiueness of the maximum if it exists. The **MLE** has the usual properties,

$$\sqrt{n}(\hat{\beta}_{ML} - \beta_0) \xrightarrow{d} \mathcal{N}(0, E[\exp(X_i'\beta_0)X_iX_i']^{-1})$$

And as usual, **inference** can be performed using the **LR, W, and LM tests**.

3.2.2 Overdispersion

The Poisson model imposes equidispersion which is very restrictive.

> **定义 3.2.1.** ***Equidispersion*** *means the conditional expectation is equal to the variance.*
> $E[Y_i|X_i] = \text{Var}(Y_i|X_i)$
> *While* ***overdispersion*** *means that the variance is bigger than the conditional mean.*
> $\text{Var}(Y_i|X_i) > E[Y_i|X_i]$

There are many causes for overdispersion: measurement error, misspecification of the condition mean, neglected heterogeneity (random parameter variation). Focusing on the neglected heterogeneity issue, assuming

$$
\begin{aligned}
E[Y_i|X_i, \epsilon_i] &= \exp(X_i'\beta_0 + \epsilon_i) \quad \epsilon_i \text{ can be any possibilities of error} \\
E[\exp(\epsilon_i)|X_i] &= 1 \\
\text{Var}(\exp(\epsilon_i)|X_i) &= \sigma^2
\end{aligned}
$$

In this case,

$$E[Y_i|X_i] = E[\lambda(X_i, \beta_0)|X_i] = E_\epsilon[\exp(X_i'\beta_0 + \epsilon_i)|X_i] = \exp(X_i'\beta_0)$$

Therefore, this sort of neglected heterogeneity **does not change the form of the conditional expectation of** Y_i.

定理 3.2.2. ***Gourieroux, Monfort and Trognon (1984)***
*If $E[Y_i|X_i] = \lambda(X_i', \beta_0)$ **is correctly specified** and the likelihood function is constructed using a probability distribution which does not necessarily correspond to the true distribution of the data, but **belongs to the family of linear exponential distribution**, then the **Quasi-Maximum Likelihood** estimator is **consistent** for β_0.*

Wrong estimator $\to$ same exponential family $\to$ still consistent. The family of linear exponential distribution includes

- Poisson distribution
- Normal distribution with fixed variance $\to$ using contiuous distribution to describe a discrete distribution.
- Binomial distribution with fixed number of trials
- Gamma distribution with fixed shape parameter

Inference is done using the results presented for the Quasi-Maximum Likelihood estimator. In particular, since the Poisson pseudo-MLE is consistent in presence of this sort of misspecification, valid inference can be based on

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(0, A^{-1}BA^{-1})$$

where

$$A = E[\exp(X_i'\beta_0 X_i X_i')] \quad B = E[(Y_i - \exp(X_i'\beta_0))^2 X_i X_i']$$

Notice that

$$\begin{aligned} Var(Y_i|X_i) &= E_{\epsilon_i}[\text{Var}(Y_i|X_i, \epsilon_i)] + \text{Var}(E_{\epsilon_i}[Y_i|X_i, \epsilon_i]) \quad \text{law of total variance} \\ &= \exp(X_i'\beta_0) + \sigma^2 \exp(2X_i'\beta_0) \\ &\geq E[Y_i|X_i] = \exp(X_i'\beta_0) \end{aligned}$$

This means that the **presence of overdispersion or underdispersion** can be tested with **LM test statistic**

$$\begin{aligned} &H_0 : \sigma^2 = 0 \to \text{equidispersion} \\ &T = \sum_{i=1}^{n} \frac{(Y_i - \exp(X_i'\hat{\beta}))^2 - Y_i}{\sqrt{2\sum_{i=1}^{n} \exp(2X_i'\hat{\beta})}} \xrightarrow{d} \mathcal{N}(0, 1) \end{aligned}$$

Or a **RESET test** to test if the specificiation $E[Y_i|X_i] = \exp(X_i'\beta_0)$ is correct.

$$\begin{aligned} &H_o : \gamma_0 = 0 \to \text{correctly specified thus equidispersion} \\ &E[Y_i|X_i] = \exp(X_i'\beta_0 + \gamma_0(X_i'\beta_0)^2) \end{aligned}$$

3.2.3 Heterogeneity and the Negative Binomial Regression Model

The assumption that Y_i has a Poisson distribution coniditonal of X_i and ϵ_i with mean $\lambda_i = \exp(X_i'\beta_0 + \epsilon_i)$ leads to the **compound Poisson regression model**.

$$P(Y_i = j|X_i, \epsilon_i) = \frac{\exp[-\exp(X_i'\beta_0 + \epsilon_i)]\exp(X_i'\beta_0 + \epsilon_i)^j}{j!} \tag{34}$$

$$P(Y_i = j|X_i) = \int_{-\infty}^{+\infty} \frac{\exp[-\exp(X_i'\beta_0 + \epsilon_i)]\exp(X_i'\beta_0 + \epsilon_i)^j}{j!} g(\epsilon_i)\, d\epsilon_i \tag{35}$$

where $g(\epsilon_i)$ is the density function of ϵ_i and we assume that X_i and ϵ_i are independent. This model can be made operational in different ways:

- Pseudo maximum likelihood estimation $\rightarrow$ Refer to previous section.
- Parametric estimation for specified $g(\epsilon_i)$
- Semiparametric estimation of β_0 and $g(\epsilon_i)$

Now focusing on the second method when $g(\epsilon_i)$ **is specified**. **MLE** can be obtained, btu the estimator **may not be robust** to departures form the additional distributional assumptions. Assume that $\exp(\epsilon_i) \sim \Gamma(\sigma^{-2}, \sigma^2)$, then $P(Y_i = j|X_i)$ is given by the **Negative-Binomial model**, Cameron and Trivedi (1986). The Poisson model is obtained as a limiting case when $\sigma^2 \rightarrow 0$, but $H_0 : \sigma^2 = 0$ cannot be tested with a standard LR or Wald test.

If the model is **misspecified** but $E[Y_i|X_i] = \exp(X_i'\beta_0)$ is correct and σ^{-2} is fixed, the **negative-binomial Pseudo-MLE** estimator is **consistent** for β_0. This follows the result from **Gourieroux, Monfort and Trognon (1984)** and the fact that the negative-binomial distribution with σ^{-2} fixed is a member of the family of linear exponential distributions.

引理 3.2.3. *Building on top of the original* ***Gourieroux, Monfort and Trognon (1984)***
If the model

- *is correct* $\xrightarrow{MLE}$ *consistent*
- *is incorrect but fix* $\sigma^{-2} \xrightarrow{MLE} \hat{\beta}$ *consistent*
- *is incorrect* $\xrightarrow{MLE} \hat{\beta}, \sigma^2$ *inconsistent*

The **score test** for $H_0 : \sigma^2 = 0$ is the overdispersion test discussed previously. Other parametric alternatives to the Poisson regression are available.

Now focusing on the third method with **semiparametric estimation** which assumes that ϵ has a discrete dsitribution with Q support points $\alpha_1, \dots, \alpha_Q$ and corresponding possibilities $\pi_1, \dots, \pi_Q$, leading to

$$P(Y_i = j|X_i) = \sum_{q=1}^{Q} \frac{\exp[-\exp(X_i'\beta_0 + \alpha_q)] \exp(X_i'\beta_0 + \alpha_q)^j}{j!} \pi_q \tag{36}$$

For a given Q, estimation of $\beta, \alpha_1, \dots, \alpha_Q, \pi_1, \dots, \pi_{Q-1}$ can be performed by **ML**. This leads to a **consistent** estimator if Q is allowed to increase at an appropriate rate. In practice, the value of Q has to be chosen, for example using information criterion. But **inferece is complciated** by the fact that the number of parameters is not fixed. This method is nice because the big integral in (35) now becomes $\sum_{q=1}^{Q} \pi_q = 1$ since ϵ_i is discrete.

3.2.4 Hurdle and Zero-Inflated Poisson Model

In some cases, the population may be contaminated by individuals for which $Y_i \equiv 0$. There are two ways to model this type of data

1. Zero-Inflated Poisson Model
2. Hurdle Model

In the **Zero-Inflated Poisson Model**, the zero outcomec can arise fomr one of two regimes. In one regime, the outcome is always zero. In the other, the ususal Poisson process is at work. Let Z_i be a Bernoulli random variable such that

$$Z_i = \begin{cases} 0 & \text{with } P(Z_i = 0|X_i) = p_1 \\ 1 & \text{with } P(Z_i = 1|X_i) = 1 - p_i \end{cases}$$

Let $P(Y_i = j|X_i, Z_i = 1) = \pi_i(j; \beta_0)$, $j = 0, 1, \dots$ be the Poisson probability function. And let $P(Y_i = 0|X_i, Z_i = 0) = 1$. Note that,

$$\begin{aligned} P(Y_i = 0|X_i) \quad & \text{law of total probability} \\ &= P(Z_i = 0|X_i)P(Y_i = 0|X_i, Z_i = 0) + P(Z_i = 1|X_i)P(Y_i = 0|X_i, Z_i = 1) \\ &= P(Z_i = 0|X_i) + P(Z_i = 1|X_i)P(Y_i = 0|X_i, Z_i = 1) \\ &= p_i + (1 - p_i)\pi_i(0; \beta_0) \end{aligned}$$

For $j > 0$,

$$\begin{aligned} P(Y_i = j|X_i) \quad & \text{law of total probability} \\ &= \underbrace{P(Z_i = 0|X_i)}_{=p_i} \underbrace{P(Y_i = j|X_i, Z_i = 0)}_{=0 \text{ because all the mass are at 0 so cannot be j}} + \underbrace{P(Z_i = 1|X_i)}_{=(1-p_i)} \underbrace{P(Y_i = j|X_i, Z_i = 1)}_{\pi(j;\beta_0)} \\ &= (1 - p_i)\pi(j; \beta_0) \end{aligned}$$

Also notice that,

$$
\begin{aligned}
E[Y_i|X_i] &= \sum_{j=0}^{\infty} jP(Y_i = j|X_i) = 0 \times P(Y_i = 0|X_i) + \sum_{j=1}^{\infty} jP(Y_i = j|X_i) \\
&= \sum_{j=1}^{\infty} j(1-p_i)\pi(j;\beta_0) \\
&= (1-p_i) \underbrace{\sum_{j=1}^{\infty} j\pi(j;\beta_0)}_{\exp(X_i'\beta_0) \text{ mean of the Poisson random variable}}
\end{aligned}
$$

Therefore, the standard **pseudo maximum likelihood does not hold here if p_i depends on X_i**. The log-likelihood function this model is written as

$$
\log L(\beta) = \sum_{i=1}^{n} \log\{[p_i + (1-p_i)\pi_i(0;\beta)]^{1(Y_i=0)} \times [(1-p_i)\pi_i(Y_i;\beta)]^{1(Y_i>0)}\} \tag{37}
$$

The **Hurdle Model** is a different extention of the basic count data model is obtained by letting the zero and positive observations be generated by different mechanisms. A binary probability model determines whether a zero or a nonzero outcome occurs, then in the latter case, we observe always a positive integer $1, 2, \ldots$ Consider a Bernoulli random variable

$$
W_i = \begin{cases} 1 & \text{with } P(W_i = 1|X_i) = 1 - q_i \\ 0 & \text{with } P(Z_i = 0|X_i) = q_i \end{cases}
$$

where q_i **may depend on X_i**.

$$
\begin{aligned}
&P(Y_i = 0|X_i, W_i = 0) = 1 \\
&P(Y_i = 0|X_i, W_i = 1) = 0 \\
&P(Y_i = j|X_i, W_i = 1) = \pi^*(j;\beta_0) \quad j = 1, 2, \ldots
\end{aligned}
$$

As the model only takes strictly positive values, $\pi^*(j;\beta_0)$ cannot be Poisson. In this case,

$$
\begin{aligned}
P(Y_i = 0|X_i) &= P(W_i = 0|X_i)P(Y_i = 0|X_i, W_i = 0) + P(W_i = 1|X_i)P(Y_i = 0|X_i, W_i = 1) \\
&= q_i \times 1 + (1-q_i) \times 0 = q_i
\end{aligned}
$$

For $j = 1, 2, \ldots$

$$
\begin{aligned}
P(Y_i = j|X_i) &= P(W_i = 0|X_i)P(Y_i = j|X_i, W_i = 0) + P(W_i = 1|X_i)P(Y_i = j|X_i, W_i = 1) \\
&= P(W_i = 1|X_i)P(Y_i = j|X_i, W_i = 1) \\
&= q_i \times 0 + (1-q_i)\pi^*(j;\beta_0)
\end{aligned}
$$

In this case, we have

$$
\begin{aligned}
E[Y_i|X_i] &= \sum_{j=0}^{\infty} jP(Y_i = j|X_i) = 0 \times P(Y_i = 0|X_i) + \sum_{j=1}^{\infty} jP(Y_i = j|X_i) \\
&= \sum_{j=1}^{\infty} j(1-p_i)\pi^*(j;\beta_0) \\
&= (1-p_i) \underbrace{\sum_{j=1}^{\infty} j\pi^*(j;\beta_0)}_{\text{mean of } \pi^*(j;\beta_0) \text{ cannot be } \exp(X_i'\beta_0)}
\end{aligned}
$$

Again, the **standard pseudo maximun likelihood result does not hold here.** Then, the log-likelihood function has the form

$$
\log L(\beta) = \sum_{i=1}^{n} \{1(Y_i = 0)(\log q_i) + 1(Y_i > 0)\log(1-q_i) + 1(Y_i > 0)\log[\pi^*(Y_i;\beta)]\} \tag{38}
$$

Notice that this function is separable meaning the **correlated unobserved heterogeneity can be allowed for** and integrated out numerically. Can model q_i's separately, $q_i = \frac{\exp(X_i'\gamma_0)}{1+\exp(X_i'\gamma_0)}$ with this then we can estiamte the β's.

Usually, $\pi^*(j;\beta_0)$ is specified as a truncated Poisson of the form

$$
\pi^*(j;\beta_0) = \frac{\exp(-\lambda_i)\lambda_i^j}{(1-\exp(-\lambda_i))j!}, \quad j > 0
$$

with $\lambda_i = \exp(X_i'\beta_0)$. However, there is no real truncation in this model, so an equally valid specification is

$$
\pi^*(j;\beta_0) = \frac{\exp(-\lambda_i)\lambda_i^{j-1}}{(j-1)!}, \quad j > 0
$$

When the **truncated Poisson specification is used**, we can test the **null of no hurdle** through

$$
\begin{aligned}
&H_0 : \beta_0 = \gamma_0 \\
&q_i = \exp(-\exp(X_i'\gamma_0))
\end{aligned}
$$

In any case, **consistency depends on the distributional assumptions.**

第 4 章 Limited Dependent Variable Models

In some applications, the domain of the variate of interest is limited. In application, the bounds of the domain of Y often have to be taken into account. Traditionally, there are three forms of limited dependent variable models:

1. Truncated data
2. Censored data
3. Sample selection which is also called incidental truncation

4.1 Truncated data

定义 4.1.1. *A sample Y is said to come from a **truncated distribution** when it is not possible to obtain observations from part of the domain of Y.*

If Y is a **continuous random variable** with pdf $f(y)$ and a is a constant, we have **left-truncation at a** if the sample is drawn from the conditional distribution

$$f(y|Y>a)=\frac{f(y)}{P(Y>a)}$$

We have also truncation if the regressors are also not observed when $Y<a$. For a **discrete random variable** with support on $0,1,\ldots,\infty$, we have **left-truncation at a** if the sample is drawn from

$$P(Y=j|Y>a)=\frac{P(Y=j)}{1-\sum_{k=0}^{\lfloor a\rfloor}P(Y=k)'}$$

Notice that truncation is only problematic if it depends on Y or on another variable that is not independent of Y and that is not used as a regressor. Under certain conditions, it is possible to perform inference about the entire population using truncated samples. The problem is that this inference tends to be **sensitive to distributional assumptions**.

Example

Normal distribution with mean μ and variance σ^2

$$E[Y_i|Y_i>a]=\mu+\sigma\underbrace{\frac{\phi(\frac{1-\mu}{\sigma})}{1-\Phi(\frac{a-\mu}{\sigma})}}_{\text{extra term}}$$

If you try to make inference on μ, it would be biased because of this extra term.

For the case of count data with $E[Y_i|X_i] = \exp(X_i'\beta_0)$

$$E[Y_i|X_i, Y_i > a] = \frac{\exp(X_i'\beta_0)}{1 - \underbrace{\sum_{k=0}^{\lfloor a \rfloor} P(Y_i = k|X_i)}_{\text{notice here}}}$$

The result depends on the probability function of the data.

In both of the examples, the expectation for the truncated data **depends on the shape of the conditional distribution**. Because of this, the estimation is often performed by **maximum likelihood**. Assume we know the distribution of the data.

4.1.1 Truncated normal regression model

The **Truncated normal regression model** is defined by

$$Y_i^* = X_i'\beta_0 + u_i^*, \quad u_i^* \sim \mathcal{N}(0, \sigma^2)$$

$$(Y_i, X_i) = \begin{cases} (Y_i^*, X_i) & \text{if } Y_i^* > 0 \\ \text{non-observable} & \text{if } Y_i^* \leq 0 \end{cases}$$

This implies that

$$E[Y_i|X_i, Y_i > 0] = X_i'\beta_0 + \sigma \frac{\phi(\frac{-X_i'\beta_0}{\sigma})}{1 - \Phi(\frac{-X_i'\beta_0}{\sigma})}$$

Again with the extra term, the OLS assumption of $E[Y_i|X_i] = X_i'\beta_0$ is violated $\rightarrow$ use MLE. The parameters of interest can be estimated from the likelihood function

$$L(\beta, \sigma) = \prod_{i=1}^{n} \frac{1}{\sigma} \frac{\phi(\frac{Y_i - X_i'\beta}{\sigma})}{1 - \Phi(\frac{-X_i'\beta_0}{\sigma})}$$

4.2 Censored data

Censoring is a problem of partial observability. For example, if Y^* is a random variable and a is a constant, we have **left-censoring at a** if we can only observe

$$Y = \max\{Y^*, a\} = \begin{cases} a & \text{if } Y^* \leq a \\ Y^* & \text{if } Y^* > a \end{cases}$$

We can also have **right truncation**, in this case, $Y = \min\{Y^*, a\}$.

Focusing on left-censored data. If Y^* is a **discrete random variable** with support on $0, 1, \ldots, \infty$, the probability function of Y is given by

$$P(Y = j) = \begin{cases} 0 & \text{if } j < a \\ P(Y^* \leq a) & \text{if } j = a \\ P(Y^* = j) & \text{if } j > a \end{cases}$$

If Y^* is a **continuous random variable** with pdf $f(y^*)$, then Y has the **mixed distribution**

$$P(Y \leq k) = \begin{cases} 0 & \text{if } k < a \\ P(Y^* \leq k) & \text{if } k \geq a \end{cases}$$

It is assumed that the **regressors are fully observed**, even for cases with censored Y^*.

4.2.1 Tobit model

The standard regression model for continuous censored data is the **Tobit** which was proposed by Tobin (1958). The model is defined by

$$Y_i^* = X_i'\beta_0 + u_i^*, \quad u_i^* \sim \mathcal{N}(0, \sigma^2)$$
$$Y_i = \begin{cases} 0 & \text{if } Y_i^* \leq 0 \\ Y_i & \text{if } Y_i^* > 0 \end{cases}$$

The model is appropriate if we are interested in **making inference about Y_i^* not Y_i**. In this case,

$$E[Y_i|X_i] = X_i'\beta_0 \Phi(\frac{X_i'\beta_0}{\sigma}) + \sigma\phi(\frac{X_i'\beta_0}{\sigma})$$

ML inference is based on the likelihood function

$$L(\beta, \sigma) = \prod_{i=1}^{n} \left[1 - \Phi(\frac{X_i'\beta_0}{\sigma})\right]^{1(Y_i=0)} \left[\frac{1}{\sigma}\phi\left(\frac{y_i - X_i'\beta}{\sigma}\right)\right]^{1(Y_i>0)}$$